

On the Impact of Pruning on Model Robustness

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Introduction

1 Background

Large neural networks have produced state-of-the-art results in numerous fields such as computer vision or natural language processing. To deploy these neural network model in an industry application, the most common option is training the model in a data center utilizing huge GPU clusters, liquid cooling, etc. At inference time, the clients could also access the model in the data center via internet.

However, with the proliferation of Internet-of-Things (IoT) devices and applications, neural network models are required to run more efficiently on resource-constrained devices to facilitate the growing number of edge AI use cases [1]. Efficiency or compactness refers to the model giving outputs accurate enough to enable the specific application, while also inducing low memory and energy-consumption overhead. The latter two translates to the model having a small parameter count and an architecture which minimizes latency and floating-point operations (FLOPs). Depending on the deployment scenario, a model may receive inputs that are far from the distribution of the training data or inputs that are manipulated to make the model produce a wrong output. Robustness of the model refers to its ability to maintain its performance even on such inputs. To sum up, a trained neural network has to satisfy accuracy, compactness and robustness requirements in order to become qualified for industrial applications.

In academia, there has been extensive research to improve them, but most works tackled them separately [2] [3]. To improve accuracy, the general approach was training over-parametrized networks with regularization. Model compression techniques (i.e. weight pruning, knowledge distillation) and novel architecture have been proposed to achieve compactness while retaining accuracy. Finally, increasing robustness has been achieved by augmenting the training dataset to better represent the distribution of inputs at inference [4].

However, meeting these requirements simultaneously - so not increasing one at the expense of another - was found to be a challenge [5]. Therefore, it is important to evaluate a technique tackling one of the three requirements on all three aspects. In my case, measuring the impact of model compression on accuracy and robustness as well. In the following chapters, I give an overview of the techniques relevant to this work.

1.1 Global Magnitude-based Pruning

There are many contemporary methods for model compression. The main four categories include network pruning, knowledge distillation, tensor decomposition, and quantization [6], although it is also common to propose hybrid strategies. Network pruning refers to evaluating weights or groups of weights of the model based on a certain importance criteria and removing the ones which were considered the least important. This importance signal could weight magnitude, the sensitivity of loss with respect to the weight, or other heuristics. After pruning, a fine-tuning or retraining phase is applied using the remaining weights in order to refit the pruned model to the training data. When the magnitude is utilized for representing the importance of weights, the pruning step uses a threshold for comparison, and every weight with magnitude below the threshold gets removed. Global pruning refers

to having only one threshold and using it across all layers, regardless of their shape and function (i.e. classifier layer, convolutional layer). If every layer has its separate, local threshold, the pruning is applied uniformly across the entire network, which can lead to sub-optimal sparsity and the increased complexity of tuning these thresholds for a deep network [3]. In contrast, global pruning enables layers with different roles (i.e. feature extractor, classifier) to contribute to the overall sparsity non-uniformly.

1.2 Out-of-Distribution and Adversarial Robustness

A basic assumption in traditional machine learning theory is that training and test samples follow the same distribution. However, in many applications this assumption may be violated due to distributional shifts in the test set. Although the degraded performance due to the distributional shifts may not be critical in recommendation or image classification systems, decreased accuracy in healthcare or self-driving cars could even result in fatalities. Therefore, the Out-of-Distribution (OOD) Generalization Problem has been formulated and an increasing number of methods and evaluation procedures have been developed accordingly [7]. Popular categorization distinguishes shifts as either covariate or concept shifts. The former refers to change in the input features distribution $P(x)$, while the latter indicates a different underlying concept or label mapping $P(Y|X)$ present compared to the one generating the training samples.

Besides OOD inputs, an application-ready model is required to be robust to so-called adversarial inputs as well. By definition, adversarial inputs are maliciously altered data points of the original distribution which are used to make the model generate incorrect outputs [8]. As the name implies, there are malevolent actors involved who apply sophisticated methods to hide these modifications from a human inspector. In case of vision models, various optimization strategies can be utilized to find a specific perturbation applied to the input image to achieve its misclassification. These strategies for corrupting in-distribution inputs are referred to as adversarial attacks, while adversarial training indicates certain modifications in the training process to increase robustness to such inputs.

2 Related Works

papers in bookmarks

3 Methodology

In this section I present the benchmark dataset and adversarial algorithm utilized to evaluate OOD and adversarial robustness of a (pruned) network.

3.1 CIFAR Corrupted Benchmark

The CIFAR-10 and CIFAR-100 datasets have been used as benchmarks for decades in the field of computer vision [9]. The two datasets consist of 60,000 32x32 RGB images with 10 and 100 different classes, respectively. The images are uniformly distributed across the classes

and objects are at the center, thus it serves as a clean baseline without any modification that may happen in real-world scenarios.

To facilitate the prediction of the impact of common image corruptions the trained model, the CIFAR-10-C and CIFAR-100-C datasets were constructed [10]. The authors applied 19 different types of corruptions at varying intensities on the test sets of CIFAR-10 and CIFAR-100, respectively. More specifically, corruptions are categorized into following categories: noise, blur, weather conditions, and digital corruptions, while intensity levels span from 1 to 5. Figure 1 shows an example for every category, applied on the same image from CIFAR-10 dataset.



Figure 1: Different types of corruptions with severity level 5 on a CIFAR-10 32x32 RGB image

In the context of the OOD generalization problem mentioned in section 1.2, these real-world distortions cause a covariate shift in the distribution of the test data. In other words, only input features suffer a distributional shift, while the underlying concept - the class of photographed object in this case - remains the same.

3.2 Fast Adaptive Boundary

Apart from OOD robustness, a neural network may need be prepared for adversarial attacks, which were introduced in section 1.2. These adversarial attacks can be categorized in many ways. Methods which have full control over the network are called white-box attacks, while black-box attacks can only give an input and inspect the corresponding output of the model. In case of the former category, some algorithms are looking for the minimal perturbation on the input image which achieves misclassification, while others return an altered image whose distance from the original is just thresholded. Finding minimal perturbations for deep neural networks yields NP-hard optimization problems whose computational costs are prohibitive. Thus, finding scalable heuristics which find high-quality results fast became the focus of current literature. Regarding adversarial robustness of trained models, an attack that overestimates the minimal perturbation also overestimates the robustness of the model, thus encouraging false conclusions. The Fast Adaptive Boundary (FAB) attack was specifically tailored to yield high-quality results fast with minimal hyperparameter-tuning required [11].

3.3 Fourier Sensitivity Method

4 Experiments

mention architecture, include description, dataset is cifar10 and cifar10c. fab was run short to limit time useage

4.1 Sparsity-Robustness tradeoff

4.2 Sensitivity-analysis

5 Summary

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A Appendix

Table 1: Real-World Mapping of CIFAR-10-C / CIFAR-100-C Corruptions

Category	Corruption Type	Real-World Operational Scenario
Noise	Gaussian Noise	Low-light environments creating electrical sensor thermal noise.
	Shot Noise	Quantum photon fluctuations when capturing images under low light.
	Impulse Noise	Bit errors during digital transmission or camera sensor malfunction.
Blur	Defocus Blur	Camera lens being out of focus relative to the target object.
	Glass Blur	Camera lens smeared with oil, smudges, or viewing through frosted windows.
	Motion Blur	High-speed movement of the camera tracking system or fast-moving targets.
	Zoom Blur	Rapid forward movement toward an object or active focal length adjustments.
Weather	Snow	Severe cold weather obstruction altering textures and scattering light.
	Frost	Ice crystals forming over outdoor security or automotive camera lenses.
	Fog	Atmospheric light scattering reducing contrast and detail over distances.
	Brightness	Intense direct sunlight, glare, or high-beam headlights facing the sensor.
Digital	Contrast	Sudden drop in lighting or bad automatic exposure algorithms.
	Elastic Trans.	Physical camera vibrations or ripples from looking through heat shimmer waves.
	Pixelate	Downsampling, low resolution crop, or low-bandwidth emergency stream degradation.
	JPEG Comp.	Visual compression artifacts from saving or transmitting files over the web.